Xinjia Zhao

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RESEARCH INTERESTS

Nanopore single-molecule sensing and bioanalysis. Proposed the PIToN model for engineering aerolysin pores, achieving sub-dalton single-molecule discrimination of protein citrullination; developed new assays for neutral glycans, sialic-acid linkage isomers, and cisplatin–DNA adducts. Actively controlled, repeatedly readable single-molecule platforms (molecular ping-pong). Mechanisms resolved by molecular dynamics combined with finite-element simulation, linking pore chemistry to ionic-current signatures.

EDUCATION

2015 – 2021 Ph.D., Condensed Matter Physics — Institute of Physics, CAS (advisor: Prof. Xinghua Lu)
2011 – 2015 B.S., Physics — Jilin University, School of Physics

APPOINTMENTS

2023 – present Research Assistant Professor — Dalian Institute of Chemical Physics, CAS (group of Prof. Guangyan Qing)
2021 – 2023 Postdoctoral Researcher — Institute of Physics, CAS (advisor: Prof. Xinghua Lu)

HONORS AND AWARDS

- **2025** First Prize, Science and Technology Award of the China Association for Instrumental Analysis (CAIA)
- **2024** Liaoning "Xingliao Yingcai" Postdoctoral Reserve Program
- **2023** High-level Talent (Young Talents) Program, DICP

RESEARCH FUNDING

- **2024 – 2025** Principal Investigator, DICP–First Affiliated Hospital of Dalian Medical University Joint Fund (DMU-1&DICP UN202307): polymer-modified porous titanium alloy scaffolds for large segmental bone defect repair (¥500,000)
- **2018 – 2020** Participant, NSFC Major Research Plan (91753136): nanopore detection of dynamic modifications of biomacromolecules (¥700,000)
- **2016 – 2019** Participant, NSFC General Program (31571012): nanopore sequencing with DNA self-assembly labels (¥660,000)

PUBLICATIONS

*24 published papers and 1 manuscript in press; # co-first author, * corresponding author.*

[1] Xiaoyu Chen, Jiao-ao Sun, **Xinjia Zhao**, Zhuang Mi, Haitao Tang, Kaikai Chen, Xinyan Shan, Xinghua Lu*. Quartz-supported suspended membrane for ultra-sensitive solid-state nanopore with sub-1 pF capacitance. *Adv. Funct. Mater.*, **2026**, 36, e77522.

[2] Dongdong Wang, **Xinjia Zhao***, Minmin Li, Jiarong Zhang, Yingying Lian, Guangyan Qing*. Multiscale theoretical calculations drive mechanism analysis and application expansion of nanopore detection. *Anal. Sens.*, **2026**, 6, e202600002.

[3] Jiaqi Li, **Xinjia Zhao**, Minmin Li, Wei Yu, Yuting Li, Chen Zhang, Hongzhi Cao, Kan Zhong, Jing Wang, Yuchen Cao, Xiaonong Li, Han Zhou, Guoxiong Wang, Xinmiao Liang*, Guangyan Qing*. Single-feature identification of α2–8 linked sialoglycans using engineered aerolysin nanopores: A paradigm for glycan linkage analysis. *J. Am. Chem. Soc.*, **2026**, 148, 10649–10662.

[4] Jing Wang, Minmin Li*, Chen Zhang, **Xinjia Zhao**, Yuting Xiong, Yuchen Cao, Dongdong Wang, Xiaonong Li, Xinmiao Liang*, Guangyan Qing*. Identification of isomerically diverse ginsenosides using engineered aerolysin nanopore via non-translocation blockade sensing. *Angew. Chem. Int. Ed.*, **2025**, 64, e202506741.

[5] Wenjing Sun, Xinmiao Zhao, **Xinjia Zhao**, Lingkai Meng, Mingliang Tang, Jiaqi Li, Yongxin Chang, Yuting Xiong, Hao Wang, Jinghua Chen*, Guangyan Qing*. Significantly enhanced capture efficiency of cell-imprinted material for circulating tumor cells via a flexible and ultra-strong double-armed phenylboronic acid design. *Biomaterials*, **2025**, 322, 123397.

[6] **Xinjia Zhao**, Yahui Zhang, Guangyan Qing*. Nanopore toward genuine single-molecule sensing: Molecular ping-pong technology. *Nano Lett.*, **2025**, 25, 3692–3706.

[7] Yahui Zhang, **Xinjia Zhao**, Guangyan Qing*. Smart polymers advance analytical chemistry. *Trends Anal. Chem.*, **2025**, 187, 118198.

[8] Yongxin Chang, Juan Shao, **Xinjia Zhao**, Haijuan Qin, Yanqing Du, Junrong Li, Qiongya Li, Wenjing Sun, Guoxiong Wang, Guangyan Qing*. Precise AIE-based ternary co-assembly for saccharide recognition and classification. *Adv. Sci.*, **2024**, 11, 2405613.

[9] Minmin Li, Jing Wang, Chen Zhang, **Xinjia Zhao**, Yuting Xiong, Yuchen Cao, Dongdong Wang, Xiaonong Li, Xinmiao Liang*, Guangyan Qing*. Single-molecule identification and quantification of steviol glycosides with a deep learning-powered nanopore sensor. *ACS Nano*, **2024**, 18, 25155–25169.

[10] Zhuang Mi, **Xinjia Zhao**, Xiaoyu Chen, Yichao Wang, Xinyan Shan*, Kaikai Chen*, Xinghua Lu*. Ten thousand recaptures of a single DNA molecule in a nanopore and variance reduction of translocation characteristics. *ACS Nano*, **2024**, 18, 23243–23252.

[11] Zhuang Mi, Xiaoyu Chen, **Xinjia Zhao**, Haitao Tang, Wenyu Wang, Xinyan Shan, Xinghua Lu*. High-precision high-speed nanopore ping-pong control system based on field programmable gate array. *Rev. Sci. Instrum.*, **2024**, 95, 073202.

[12] Wenqi Lu#, **Xinjia Zhao#**, Minmin Li, Yuting Li, Chen Zhang, Yuting Xiong, Jiaqi Li, Han Zhou, Xianlong Ye, Xiaonong Li, Jing Wang, Xinmiao Liang*, Guangyan Qing*. Precise structural analysis of neutral glycans using aerolysin mutant T240R nanopore. *ACS Nano*, **2024**, 18, 12412–12426.

[13] Wenjing Sun, Xin You, **Xinjia Zhao**, Xiaoyu Zhang, Chunhui Yang, Fusheng Zhang, Jiaqi Yu, Kaiguang Yang, Jixia Wang, Fangfang Xu, Yongxin Chang, Boxin Qu, Xinmiao Zhao, Yuxuan He, Qi Wang*, Jinghua Chen*, Guangyan Qing*. Precise capture and dynamic release of circulating liver cancer cells with dual-histidine-based cell imprinted hydrogels. *Adv. Mater.***, 2024**, 36, e2402379.

[14] Jie Xiao, Muyu Cong, Minmin Li, Xin Zhang, Yahui Zhang, **Xinjia Zhao**, Wenqi Lu, Zhimou Guo, Xinmiao Liang*, Guangyan Qing*. Self-assembled nanoporous metal–organic framework monolayer film for osmotic energy harvesting. *Adv. Funct. Mater.***, 2024**, 34, 2307996.

[15] **Xinjia Zhao**, Haijuan Qin, Mingliang Tang, Xiaoyu Zhang, Guangyan Qing*. Nanopore: Emerging for detecting protein post-translational modifications. *Trends Anal. Chem.***, 2024**, 173, 117658.

[16] Yuting Xiong#, Minmin Li#, Yuchen Cao, Zan Li, Yongxin Chang, **Xinjia Zhao**, Guangyan Qing*. Nanofluidic device for detection of lysine methylpeptides and sensing of lysine methylation. *Anal. Chem.***, 2023**, 95, 7761–7769.

[17] Yue Qin#, Xiaoyu Zhang#, Yanling Song, Bowen Zhong, Lu Liu, Dongdong Wang, Yahui Zhang, Wenqi Lu, **Xinjia Zhao**, Zhiqi Jia, Minmin Li, Lihua Zhang*, Guangyan Qing*. A highly sensitive nanochannel device for the detection of SUMO1 peptides. *Chem. Sci.***, 2023**, 14, 8360–8368.

[18] Hao Dong, Yueyi Wang, Xiangqian Tang, **Xinjia Zhao**, Yang An, Wenyu Wang, Jianmei Li, Xinyan Shan, Xinghua Lu*. Geometrical and magnetic properties of small titanium and chromium clusters on monolayer hexagonal boron nitride. *Phys. Chem. Chem. Phys.***, 2023**, 25, 6079–6088.

[19] Yikun Yao, **Xinjia Zhao**, Xiangqian Tang, Jianmei Li, Xinyan Shan*, Xinghua Lu*. Laser etching of 2D materials with single-layer precision up to ten layers. *J. Laser Appl.***, 2022**, 34, 042051.

[20] Yang An, Xiangqian Tang, Wenyu Wang, Hao Dong, **Xinjia Zhao**, Muyu Wang, Ye Xia, Xinyan Shan*, Xinghua Lu*. Spatially resolved stimulation for the controlled debromination in single molecules on a surface. *ACS Nano***, 2022**, 16, 18592–18600.

[21] Xiaoyu Chen, **Xinjia Zhao**, Ruiping Ma, Ying Hu, Chengjun Cui, Zhuang Mi, Ruifen Dou, Dun Pan, Xinyan Shan*, Lihua Wang, Chunhai Fan, Xinghua Lu*. Ionic current fluctuation and orientation of tetrahedral DNA nanostructures in a solid-state nanopore. *Small***, 2022**, 18, 2107237.

[22] **Xinjia Zhao**#, Yuru Liu#, Xiaoyu Chen, Zhuang Mi, Wei Li, Pengye Wang*, Xinyan Shan*, Xinghua Lu*. Detection and characterization of single cisplatin adducts on DNA by nanopore sequencing. *ACS Omega***, 2021**, 6, 17027–17034.

[23] **Xinjia Zhao**, Xiaoyu Chen, Zhuang Mi, Yuru Liu, Wei Li, Xinyan Shan*, Xinghua Lu*. Revealing differential interaction forces during nanopore DNA sequencing. *J. Phys. Chem. B***, 2021**, 125, 5045–5051.

[24] **Xinjia Zhao**, Ruiping Ma, Ying Hu, Xiaoyu Chen, Ruifen Dou, Ke Liu, Chengjun Cui, Huajie Liu, Qian Li, Dun Pan, Xinyan Shan*, Lihua Wang*, Chunhai Fan, Xinghua Lu*. Translocation of tetrahedral DNA nanostructures through a solid-state nanopore. *Nanoscale***, 2019**, 11, 6263–6269.

[25] **Xinjia Zhao**#, Jiarong Zhang#, Juanjuan Zhao#, Wei Yong, Zeyuan Zhao, Minmin Li, Yuchen Cao, Wenjing Sun, Jing Wang, Dongdong Wang, Jiaqi Li, Kuoxi Xu, Yangyang Bian, Guangyan Qing*. Sub-dalton discrimination of citrullination by a semi-rationally designed aerolysin nanopore. *Nat. Commun.***, 2026, in press**.

PATENTS (GRANTED)

[1] Mold and method for fabricating an insulating base, and use of the base in molecular detection. Patent No. ZL201811088465.1. Inventors: Xinjia Zhao, Xinyan Shan, Xinghua Lu.

[2] Force-spectroscopy analysis method and apparatus based on nanopore sequencing. Patent No. ZL202110598008.2. Inventors: Xinjia Zhao, Xiaoyu Chen, Zhuang Mi, Xinyan Shan, Xinghua Lu.

[3] Apparatus and method for controlling the motion of biomolecules in a nanopore. Patent No. ZL201911053046.9. Inventors: Xiaoyu Chen, Xinjia Zhao, Zhuang Mi, Yaru He, Xinyan Shan, Xinghua Lu.

PATENT APPLICATIONS

[4] Nanopore microfluidic chamber and its applications. Application No. CN202010947474.2. Inventors: Xiaoyu Chen, Xinjia Zhao, Zhuang Mi, Xinyan Shan, Xinghua Lu.

[5] Device and method for high-throughput screening of biomolecules. Application No. CN202010946762.6. Inventors: Xiaoyu Chen, Xinjia Zhao, Zhuang Mi, Xinyan Shan, Xinghua Lu.

PROFESSIONAL SERVICE AND MEMBERSHIPS

- **2026 – present** Full Member, Sigma Xi, The Scientific Research Honor Society (elected by nomination)
- **2024 – present** Committee member, Cardiovascular Disease Medical-Engineering Integration Committee, Liaoning Society for Cell Biology

TECHNICAL SKILLS

Solid-state nanopore fabrication and low-current ionic recording; molecular ping-pong control and FPGA-driven instrumentation; molecular dynamics and finite-element simulation of pore–molecule interactions; scientific data analysis.